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Bitcoin 2023: Keep "Hodling"?

Whether it goes up or down in the next year, or five or 10 years, I don't know. But the one thing I'm pretty sure of is that [Bitcoin]doesn't produce anything. It's got a magic to it and people have attached magic to lots of things.

—Warren Buffet, chairman and CEO, Berkshire Hathaway Inc.[1]

I think the average investor should ask himself or herself what do you have in your portfolio that has [Bitcoin's] kind of track record—number one; is very, very underpenetrated; can provide a service of insurance against financial catastrophe that no one else can provide; and can go up ten times or fifty times. The answer is: nothing.

—Bill Miller, founder, chairman, CIO, Miller Value Partners[2]

In early March 2023, John Brown, a hedge fund manager based in Miami, was reviewing his portfolio. His focus was on the fund's long-term position in bitcoin. Brown had previously initiated a small position in the cryptocurrency (crypto) at the urging of one of the fund's limited partners (LPs).[3] Although the investment, made in September 2017, had been a resounding success, increasing more than fivefold in less than six years (**Exhibit 1**), Brown had to endure striking price swings. Through thick and thin, Brown had continued to "hodl"—crypto slang meaning to buy and hold.[4]

Brown had paid around USD4,200[5] per bitcoin and in just a few months, the price had more than quadrupled, peaking at over USD19,000 in December 2017. The typically measured Brown had been ecstatic. Just one year later, the price of a bitcoin was under USD4,000, a loss of over 75% from its high. Bitcoin had another massive rally during the COVID-19 pandemic, eventually reaching USD68,000 in November 2021. In 2022, bitcoin crashed yet again, at one point falling by 80%. In late February 2023, the price of a bitcoin had rebounded a bit and was trading at around USD23,000.

[1] Tanaya Macheel, "Warren Buffett Gives His Most Expansive Explanation for Why He Doesn't Believe in Bitcoin," CNBC May 2, 2022, https://www.cnbc.com/2022/04/30/warren-buffett-gives-his-most-expansive-explanation-for-why-he-doesnt-believe-in-bitcoin.html (accessed Jul. 10, 2023).

[2] Rob Lenihan, "US Billionaire Bill Miller Bets 50% of His Wealth in Bitcoin," *The Street*, January 14, 2022, https://www.thestreet.com/investing/cryptocurrency/fund-manager-bill-miller-says-50-personal-wealth-in-bitcoin (accessed Jul. 10, 2023).

[3] For a discussion of Brown's initial investment, see Yiorgos Allayannis and Aaron Fernstrom, "Bitcoin: Investment or Illusion?," UVA-F-1819 (Charlottesville, VA: Darden Business Publishing, 2017).

[4] Some also used *hodl* in acronym form, HODL, to stand for "hold on for dear life."

[5] USD = US dollars; BTC = bitcoin.

This fictional case was prepared by Ian Appel, Associate Professor of Business Administration, and George (Yiorgos) Allayannis, Robert F. Bruner Distinguished Professor of Business Administration. It was written as a basis for class discussion rather than to illustrate effective or ineffective handling of an administrative situation. Copyright © 2023 by the University of Virginia Darden School Foundation, Charlottesville, VA. All rights reserved. *To order copies, send an email to* sales@dardenbusinesspublishing.com. *No part of this publication may be reproduced, stored in a retrieval system, used in a spreadsheet, or transmitted in any form or by any means—electronic, mechanical, photocopying, recording, or otherwise—without the permission of the Darden School Foundation.* Our goal is to publish materials of the highest quality, so please submit any errata to editorial@dardenbusinesspublishing.com.

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The dramatic decline in the price of bitcoin over the past year and a half had Brown questioning his investment. While bitcoin had recovered from similar declines in the past, he knew that there were no guarantees that this would happen again. A recovery in bitcoin's price hinged on demand from other investors. The recent runup in bitcoin's price had been fueled, in part, by demand from large investors such as hedge funds (e.g., Tudor Investment Corp.) and other entities (e.g., Tesla, Inc.). Would this demand return? Not only had the front-page news failure of prominent players in the crypto industry (e.g., FTX Trading, Ltd. [FTX]) damaged confidence, but there were also long-simmering concerns related to bitcoin's environmental footprint and use in criminal activities.

It was time for Brown to reevaluate the reasons he had purchased bitcoin in the first place and assess what, if anything, had since changed. Many of his LPs had turned bearish on bitcoin and were urging him to sell off the position. Some even thought he should short it. Bitcoin, after all, had no intrinsic value—it was not backed by other assets, did not produce cash flows, and could not be used to make anything.

The Bitcoin Blockchain

Bitcoin was created by Satoshi Nakamoto, an individual or group using a pseudonym, in a 2008 white paper.[6] While thousands of other cryptos had been created since, bitcoin was the largest, with a market cap of over USD400 billion. The second-largest crypto, ether (associated with the Ethereum blockchain), had a market cap less than half of bitcoin's. Indeed, bitcoin was larger than the next 10 cryptos combined (**Exhibit 2**). As a common saying in the crypto world went, bitcoin was king.

Ownership of a bitcoin consisted of an entry in a digital ledger (i.e., a list of credits and debits to accounts). For example, if Alice bought a bitcoin, the ledger was updated to say that Alice's account (known as a "wallet") had been credited with BTC1. A bitcoin was divisible, so if she proceeded to send BTC0.25 to Bob, Alice's balance would fall by BTC0.25 and Bob's would increase by the same amount. Bitcoin and other cryptos used a special type of digital ledger known as a blockchain. When new transactions occurred on the network, they were grouped together to form a block. This block would then be combined with (chained to) existing block to form the blockchain. The full blockchain contained a record of every transaction that had ever occurred on the network.

At first blush, a blockchain was conceptually similar to other types of digital ledgers. Consider, for example, a Venmo transaction. If Alice sent Bob USD25 via Venmo, a digital ledger would be updated to reflect that Alice's balance had decreased by USD25 and Bob's had increased by USD25. This transaction simply consisted of Venmo updating a ledger, much in the same way that the bitcoin transaction consisted of a ledger being updated. The key difference was that a Venmo transaction required a centralized entity to update the ledger. A bitcoin transaction did not.

The decentralization of bitcoin was a consequence of using a consensus mechanism known as "proof of work" to verify transactions. Put simply, when a user initiated a transaction, this information was broadcast to miners—essentially, powerful computer systems. Miners would then verify that transactions were valid and compete against each other to solve a complex mathematical problem associated with each block of transactions. The first miner to solve the problem would add a new block to the existing blockchain, earning a reward in the form of newly issued bitcoin. As of March 2023, the reward for mining a block was BTC6.25, or close to USD130,000. Rewards decreased over time, eventually converging to zero around the year 2140, at which time the supply of bitcoin would remain fixed at 21 million.[7]

[6] Satoshi Nakamoto, "Bitcoin: A Peer-to-Peer Electronic Cash System," Bitcoin.org, https://bitcoin.org/bitcoin.pdf (accessed Jul. 10, 2023).

[7] For a more detailed discussion of bitcoin mining, see Kinda Hachem, "The Economics of Cryptocurrency," UVA-GEM-0190 (Charlottesville, VA: Darden Business Publishing, 2021).

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The Institutionalization of Bitcoin

Brown's initial purchase of bitcoin in 2017 made him an early adopter, at least in terms of institutions. In the last few years, he had noticed an increased interest in bitcoin from institutional investors. Institutional trading on Coinbase, the largest US crypto exchange, had increased nearly tenfold between 2020 and 2021, from USD120 billion to USD1.14 trillion. In comparison, trades by retail investors on Coinbase totaled USD535 billion in 2021.[8] Institutional adoption was largely driven by hedge funds, including legendary investors like Paul Tudor Jones, Stan Druckenmiller, and Alan Howard.[9] A 2022 PricewaterhouseCoopers (PwC) survey of traditional (i.e., not primarily invested in crypto) hedge funds found that nearly one-third of surveyed funds invested in digital assets, particularly in bitcoin and ether.[10] Funds with positions in crypto had an average allocation of 4% of assets under management (AUM). Crypto-focused hedge funds had also experienced significant growth but remained small (in aggregate) relative to traditional funds. There had also been limited adoption by other types of asset managers that tended to employ more conservative strategies. For example, MassMutual, a life insurance company founded in 1851, purchased USD100 million of bitcoin (representing about 0.04% of its investment portfolio) in December 2020.[11]

Many institutions purchased bitcoin in response to monetary/fiscal policies adopted in the wake of the COVID-19 pandemic. One common argument was that bitcoin would serve as a hedge against inflation. For example, in October 2020, Paul Tudor Jones of Tudor Investments explained, "The reason I recommended bitcoin is because it was one of the menu of inflation trades, like gold, like TIPS breakevens,[12] like copper…and I came to the conclusion that bitcoin was going to be the best inflation trade." Jones also noted that bitcoin had appeal from a technological perspective: "Bitcoin has this enormous contingence of really, really smart and sophisticated people who believe in it. It's like investing with Steve Jobs and Apple or investing in Google early."[13]

Institutional investors also gained exposure to the space via venture investments. Prominent investors in start-ups included both traditional venture capitalists (VCs) (e.g., Andreessen Horowitz, Bain Capital, and Sequoia Capital) as well as crypto-native funds (e.g., Binance Labs, Coinbase Ventures, and Paradigm). Crypto VC investments totaled approximately USD30 billion in 2022,[14] nearly equivalent to the amount invested in biotech start-ups.[15] The LPs of crypto VC funds were often institutions (e.g., endowments and pension funds) that wanted exposure to cryptos, but could be "skittish" about directly investing in tokens.[16] With Coinbase and other crypto companies undertaking initial public offerings (IPOs) in recent years, many institutions also had exposure to the space via public equities. For example, as of the end of 2022, the largest holder of Coinbase stock was investment management firm Vanguard Group, Inc., whose position was valued at over USD800 million.[17]

[8] Paul Vigna, "Wall Street Takes Lead in Crypto Investments," *Wall Street Journal*, February 27, 2022, https://www.wsj.com/articles/wall-street-takes-lead-in-crypto-investments-11645927004 (accessed Jul. 10, 2023).

[9] Corrie Driebusch and Paul Vigna, "The Crypto Party Is Over," *Wall Street Journal*, June 18, 2022, https://www.wsj.com/articles/the-crypto-party-is-over-11655524807 (accessed Jul. 10, 2023).

[10] "4th Annual Global Crypto Hedge Fund Report 2022," PwC, Elwood Technologies, and AIMA, 2022, https://www.pwc.com/gx/en/financial-services/pdf/4th-annual-global-crypto-hedge-fund-report-june-2022.pdf (accessed Jul. 10, 2023).

[11] Olga Kharif, "169-Year-Old MassMutual Invests $100 Million in Bitcoin," Bloomberg, December 10, 2020, https://www.bloomberg.com/news/articles/2020-12-10/169-year-old-insurer-massmutual-invests-100-million-in-bitcoin (accessed Jul. 10, 2023).

[12] A Treasury inflation-protected security (TIPS) breakeven rate was the difference in TIPS and nominal bond yields with the same maturity.

[13] Yun Li, "Paul Tudor Jones Says He Likes Bitcoin Even More Now, Rally Still in the 'First Inning,'" CNBC, October 22, 2020, https://www.cnbc.com/2020/10/22/-paul-tudor-jones-says-he-likes-bitcoin-even-more-now-rally-still-in-the-first-inning.html (accessed May 2, 2023).

[14] Alex Thorn, "Crypto VC Comes Down to Earth," Galaxy Digital, January 5, 2023, https://www.galaxy.com/research/insights/2022-crypto-vc-comes-down-to-earth/ (accessed Jul. 10, 2023).

[15] Brian Gormley, "Tales from the 2022 Biotech VC Fundraising Trail," *Wall Street Journal*, December 2, 2022, https://www.wsj.com/articles/tales-from-the-2022-biotech-vc-fundraising-trail-11671667063 (accessed Jul. 10, 2023).

[16] Yuliya Chernova, "Limited Partners Seek Assurances From Crypto Venture Funds," *Wall Street Journal*, August 29, 2022, https://www.wsj.com/articles/limited-partners-seek-assurances-from-crypto-venture-funds-11661767201 (accessed Jul. 10, 2023).

[17] "Coinbase Global, Inc (COIN)," Yahoo! Finance, https://finance.yahoo.com/quote/COIN/holders (accessed Jul. 10, 2023).

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Some corporations had adopted bitcoin as a treasury asset in recent years. The first public company to purchase bitcoin was MicroStrategy Inc., a software company founded by Michael Saylor. MicroStrategy started purchasing bitcoin in the summer of 2020; by the end of 2022, it had spent close to USD4 billion to acquire 132,500 bitcoins. Saylor's logic for the purchases centered on protecting the firm's balance sheet: "We realized that cash is trash and we needed either to shrink the capital structure or move our cash into something that is going to float on the flood of liquidity and not sink under the flood of liquidity."[18] In 2022, the firm reported an impairment loss of nearly USD1.3 billion on its bitcoin holdings.[19]

Another prominent corporation that purchased bitcoin was Tesla. In early 2021, Elon Musk announced that Tesla had purchased USD1.5 billion in bitcoin, tweeting at the time that "in retrospect, it was inevitable."[20] Tesla reduced its holding by 75% the following year. Other corporations to purchase bitcoin typically had some operations in the crypto sector. Examples included Coinbase and Block (formerly Square, Inc.).[21]

The influx of institutional investors and corporations into crypto was, in part, a result of improved infrastructure. An example of such an improvement was prime brokerage services, which included financing, custody, and advanced trading options (e.g., over-the-counter transactions and routing trades to other exchanges). Prime brokers were often viewed as critical for large investors. For example, Coinbase Prime helped MicroStrategy execute its initial purchase of bitcoin (USD425 million) with minimal price impact.[22] BlackRock integrated Coinbase Prime into its Aladdin platform, portfolio-management software widely used by institutional investors, in 2022.[23]

Investment Options

Brown also needed to assess options for how to invest in bitcoin. Since his initial investment, these options had expanded considerably. For retail investors, buying bitcoin had never been easier; it could be purchased via crypto exchanges (e.g., Coinbase), online brokerages (e.g., Robinhood), popular financial apps (e.g., PayPal and Cash App), even bitcoin ATMs located throughout the United States. For institutional investors, options had similarly expanded. Brown considered several options:

Directly purchasing bitcoin

Directly purchasing bitcoin, called spot trading, was a straightforward process: investors wired money to an exchange (e.g., Coinbase) and used these funds to trade. An important difference between buying crypto and assets such as stocks was custody, which referred to how assets were stored and protected. When an investor bought a stock, a custodian such as a bank or a broker-dealer typically held it. Assets held in custody were held separately from the assets of the custodian. Thus, even if the custodian went bankrupt, the customer's assets remained safe.

[18] Jason Zweig, "'Cash Is Trash,' So Let's Bet $425 Million on Bitcoin," *Wall Street Journal*, October 30, 2020, https://www.wsj.com/articles/cash-is-trash-so-lets-bet-425-million-on-bitcoin-11604070071 (accessed Jul. 10, 2023).

[19] Jesse Coghlan, "MicroStrategy Says No Plans to Stop Trading BTC as Paper Loss Hits $1.3B," Cointelegraph, February 3, 2023, https://cointelegraph.com/news/microstrategy-says-no-plans-to-stop-trading-btc-as-paper-loss-hits-1-3b (accessed Jul. 10, 2023).

[20] Elon Musk (@elonmusk), "In retrospect, it was inevitable," Twitter, January 29, 2021, 3:22 a.m., https://twitter.com/elonmusk/status/1355068728128516101 (accessed Jul. 10, 2023).

[21] Bitcoin Treasuries website, https://bitcointreasuries.net/ (accessed Jul. 10, 2023).

[22] Brett Tejpaul, "Coinbase Is Helping Corporate Companies Diversify with Crypto," Coinbase, December 1, 2020, https://www.coinbase.com/blog/coinbase-is-helping-corporate-companies-diversify-with-crypto (accessed Jul. 10, 2023).

[23] Brett Tejpaul and Greg Tusar, "Coinbase Selected by BlackRock; Provide Aladdin Clients Access to Crypto Trading and Custody via Coinbase Prime," Coinbase, August 4, 2022, https://www.coinbase.com/blog/coinbase-selected-by-blackrock-provide-aladdin-clients-access-to-crypto-trading-and-custody-via (accessed Jul. 10, 2023).

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Traditional custodians (e.g., Bank of New York [BNY] Mellon) as well as crypto exchanges provided bitcoin custody services. In some cases, crypto exchanges were largely unregulated, potentially posing significant risk to investors. For example, Sam Bankman-Fried, the CEO of crypto exchange FTX, allegedly engaged in fraud by using customer deposits to fund risky bets, leading to a shortfall of nearly USD10 billion.[24] Both retail and institutional customers alike could not access their funds after FTX filed for bankruptcy. One hedge fund with half its assets held by FTX (approximately USD100 million) closed as a result.[25]

As an alternative, crypto investors could also have self-custody over their assets. Self-custody involved holding crypto in a wallet, a device or program that stored the private keys used to access the assets. Typically, the highest level of security involved storing keys on a device that was not connected to the internet, known as keeping assets in cold storage. Self-custody entailed its own risks; if private keys were lost or stolen, the crypto stored on the wallet could not be recovered.

Grayscale Bitcoin Trust

Another possibility Brown considered was to purchase shares in the Grayscale Bitcoin Trust (GBTC), a closed-end fund.[26] As of March 2023, it held 630 thousand bitcoins, about 3% of the total supply. Investors could purchase GBTC shares on the secondary market, similar to an exchange-traded fund (ETF). In contrast to ETFs, the pricing of closed-end fund shares could deviate from the fund's net asset value (NAV).[27] For GBTC, the magnitude of this deviation was significant (**Exhibit 3**). In the past, shares frequently traded at a premium, reaching upward of 40% for some periods. More recently, GBTC shares traded at a significant discount. Grayscale, the firm, had attempted to convert the fund to an ETF to track the price of bitcoin more closely, but the US Securities and Exchange Commission (SEC) had opposed this conversion, arguing that because bitcoin spot markets were largely unregulated, there was "not sufficient safety from market manipulation and fraud."[28] The SEC had similarly rejected applications for spot bitcoin ETFs from a number of other firms, including WisdomTree, Fidelity, and VanEck.[29]

Futures and ETFs

In December 2017, the Chicago Mercantile Exchange (CME) introduced bitcoin futures. A futures contract was an agreement to buy or sell an asset, often commodities, on a specific future date. For example, in March 2023, one could use a futures contract to enter into an agreement to buy bitcoin in July 2023. In general, the value of such a contract closely tracked the current price of bitcoin (the spot price). If the price of bitcoin increased by USD1,000 today, a contract to buy bitcoin in four months would increase by a similar amount; however, the use of futures also entailed risk because future prices could be higher than spot prices, a phenomenon known as contango.[30]

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[24] Vicky Ge Huang, Alexander Osipovich, and Patricia Kowsmann, "FTX Tapped into Customer Accounts to Fund Risky Bets, Setting Up Its Downfall" *Wall Street Journal*, November 11, 2022, https://www.wsj.com/articles/ftx-tapped-into-customer-accounts-to-fund-risky-bets-setting-up-its-downfall-11668093732 (accessed Jul. 10, 2023).

[25] Ezra Reguerra, "Hedge Fund Closes Operations after Losing Funds in FTX Exchange: Report," Cointelegraph, February 20, 2023, https://cointelegraph.com/news/hedge-fund-closes-operations-after-losing-funds-in-ftx-exchange-report (accessed Jul. 10, 2023).

[26] A closed-end fund was a type of investment company that raised funds via an IPO and used the proceeds to purchases securities.

[27] A closed-end fund traded at a premium if the market price of its shares was greater than its NAV. It traded at a discount if the market price of its shares was less than NAV.

[28] Gunjan Banerji, "SEC Rejects Grayscale Attempt to Turn Bitcoin Fund into ETF," *Wall Street Journal*, June 30, 2022, https://www.wsj.com/articles/sec-rejects-grayscale-attempt-to-turn-bitcoin-fund-into-etf-11656617147 (accessed Jul. 10, 2023).

[29] Ben Miller, "SEC Doubles Down on Decision to Reject Grayscale Spot ETF," *Financial Times*, December 16, 2022, https://www.ft.com/content/3f113f11-b5ea-4cf3-b2bb-2d9059f6b18c (accessed Jul. 10, 2023).

[30] Storage costs were one reason futures contracts could trade in contango. By way of example, suppose a firm wanted to arrange to receive 1,000 barrels of oil in six months. One way to do this would be to buy a six-month futures contract. Alternatively, the firm could buy the oil ahead of time and store it for six months. Storing oil, however, was costly because it required significant infrastructure. Futures prices reflected costs (e.g., storage) associated with carrying an asset.

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In October 2021, ProShares launched the Bitcoin Strategy ETF (BITO), which used futures to track the price of bitcoin, and had an annual expense ratio of 0.95%.[31] BITO was one of the most successful ETF launches in history, attracting over USD1 billion in two days.[32] (**Exhibit 4** plots returns for BITO and bitcoin from October 2021 to March 2023.) The SEC approved the futures-based ETF because, in contrast to bitcoin spot markets, futures markets were subject to considerable oversight.[33]

Correlated equities

A final option was to buy stocks of companies whose performance was correlated with bitcoin and crypto markets more generally. Brown considered two options: Coinbase (COIN) and Riot Platforms (RIOT). Coinbase made the bulk of its revenue from transaction fees on crypto trades. Its performance tended to be correlated with broad crypto market performance; revenues in 2022 (during "crypto winter"[34]) fell nearly 75% compared to the year earlier.[35] RIOT, a publicly traded bitcoin miner, verified blockchain transactions. Its revenue consisted primarily of rewards paid to miners that verify a block. (**Exhibit 5** compares the performance of COIN and RIOT to bitcoin over the past year.)

FUD

In crypto, FUD—an acronym for fear, uncertainty, and doubt—was pervasive. Two common sources of FUD were bitcoin's environmental impact and its use in criminal activities. Given the attention these issues had received from policymakers, other investors, and the public at large, Brown needed to determine how they impacted his investment thesis.

Environment

The effect of bitcoin mining on the environment had been a long-standing concern. In just the last few months, Brown had come across the following headlines:

- "Bitcoin Could Rival Beef or Crude Oil in Environmental Impact"[36]
- "New York Partially Banned Cryptocurrency Mining. Now Environmentalists Want More"[37]
- "Crypto Is Mostly Over. Its Carbon Emissions Are Not"[38]
- "The Real-World Costs of the Digital Race for Bitcoin"[39]

Bitcoin's environmental impact was a consequence of mining. Miners used specialized computer equipment that used large amounts of energy. As the price of bitcoin trended higher over the last several years, miners

[31] "About the Fund," ProShares, May 2, 2023, https://www.proshares.com/our-etfs/strategic/bito (accessed Jul. 10, 2023).

[32] Katherine Greifeld, "ProShares Bitcoin ETF Tops $1 Billion Assets in Just 2 Days," Bloomberg, October 20, 2021, https://www.bloomberg.com/news/articles/2021-10-20/proshares-bitcoin-etf-tops-1-billion-in-assets-in-just-two-days (accessed Jul. 10, 2023).

[33] https://www.ft.com/content/3f113f11-b5ea-4cf3-b2bb-2d9059f6b18c.

[34] *Crypto winter* was a term used to describe a prolonged period of pricing weakness in the cryptocurrency market, similar to a bear market in stocks.

[35] MacKenzie Sigalos, "Coinbase Beats on Revenue and Earnings, but Usage Continues to Decline," CNBC, February 21, 2023, https://www.cnbc.com/2023/02/21/coinbase-coin-earnings-q4-2022.html (accessed Jul. 10, 2023).

[36] Margaret Osborne, "Bitcoin Could Rival Beef or Crude Oil in Environmental Impact," *Smithsonian Magazine*, October 3, 2022, https://www.smithsonianmag.com/smart-news/bitcoin-could-rival-beef-or-crude-oil-in-environmental-impact-180980877/ (accessed Jul. 10, 2023).

[37] Marie J. French, "New York Partially Banned Cryptocurrency Mining. Now Environmentalists Want More," *Politico*, January 7, 2023, https://www.politico.com/news/2023/01/07/new-york-cryptocurrency-mining-ban-00072564 (accessed Jul. 10, 2023).

[38] Emma Harris, "Crypto Is Mostly Over. Its Carbon Emissions Are Not.," *Atlantic*, March 22, 2023, https://www.theatlantic.com/science/archive/2023/03/crypto-bitcoin-mining-carbon-emissions-climate-change-impact/673468/ (accessed Jul. 10, 2023).

[39] Gabriel J. X. Dance, "The Real-World Costs of the Digital Race for Bitcoin," *New York Times*, April 9, 2023, https://www.nytimes.com/2023/04/09/business/bitcoin-mining-electricity-pollution.html (accessed Jul. 10, 2023).

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were incentivized to expand operations, thus increasing their energy consumption. Indeed, the energy used by bitcoin miners had increased more than tenfold since 2017 (**Exhibit 6**). In May 2021, annualized consumption reached an all-time high of approximately 140 terawatt hours (TWh), but fell dramatically after China banned bitcoin mining. In the following months, miners relocated to other jurisdictions and energy usage returned to close to its previous level.[40] As of March 2023, annualized energy usage was approximately 120 TWh. To put this number in perspective, the state of Virginia, home to over eight million people, had slightly lower annual consumption of 108 TWh.[41]

The economics of mining was such that its profitability was a function of energy costs. If energy was cheap, mining was profitable. The sources of cheap energy varied substantially across geographies. For example, prior to China's ban on mining in 2021, two areas that hosted many miners were Inner Mongolia and Sichuan. Energy in Inner Mongolia was largely generated by coal plants. Sichuan mostly used hydropower, particularly during the high-river season.[42] The aggregate carbon footprint of bitcoin mining depended on the relative mix of "green" and "brown" energy sources.[43] Estimating this mix was challenging because the source of energy used by a particular miner was often not observable to outsiders. The Bitcoin Mining Council, an industry trade group, estimated that the global bitcoin mining industry's sustainable energy mix stood at 59%. This figure, however, was subject to considerable debate.[44]

The impact of mining was not confined to carbon emissions. For example, the energy consumption of some miners could stress the electric grid and impose costs on local communities.[45] One study estimated that small businesses and residents in upstate New York paid USD90 million and USD189 million, respectively, in higher annual electricity bills because of bitcoin mining.[46]

Some argued that bitcoin could have neutral or even positive environmental effects. A 2022 report on crypto mining by the White House noted that there was often not sufficient demand or transmission capacity to utilize peak levels of renewable energy.[47] For example, if the output of a hydroelectric dam exceeded demand, this energy would be curtailed (i.e., wasted) because it was not economical to store or transmit it elsewhere.[48] If renewable producers used excess energy to mine bitcoin, however, and thus earn additional revenue, this could incentivize the construction of additional capacity.

In any case, Brown was concerned about how the environmental impact of bitcoin would impact its adoption. The increased emphasis on environmental, social, and corporate governance (ESG) in recent years, particularly for institutional investors, compounded his concerns. If bitcoin had a reputation as a brown-energy asset, would this dissuade investment? Moreover, a recent upgrade for the Ethereum blockchain had drastically reduced its carbon footprint.[49] Could this be the impetus for institutions to invest in ether instead of bitcoin?

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[40] Amy Qin and Ephrat Livni, "China Cracks Down Harder on Cryptocurrency With New Ban," *New York Times*, September 24, 2021, https://www.nytimes.com/2021/09/24/business/china-cryptocurrency-bitcoin.html (accessed Jul. 10, 2023).

[41] "State of Virginia Energy Sector Risk Profile," US Department of Energy, 2016, https://www.energy.gov/sites/prod/files/2016/09/f33/VA_Energy%20Sector%20Risk%20Profile.pdf (accessed Jul. 10, 2023).

[42] Matteo Benetton, Giovanni Compiani, and Adair Morse, "When Cryptomining Comes to Town: High Electricity-Use Spillovers to the Local Economy," May 14, 2021, https://ssrn.com/abstract=3779720 (accessed Jul. 10, 2023).

[43] "Understanding the Differences between Brown Energy and Green Energy," Abest Green Power, https://abestgreenpower.com/brown-vs-green-energy/ (accessed Jul. 10, 2023).

[44] Mandy DeRoche, Jeremy Fisher, Nick Thorpe, and Megan Wachspress, "The Energy Bomb: How Proof-of-Work Cryptocurrency Mining Worsens the Climate Crisis and Harms Communities Now," Sierra Club and Earthjustice, September 2022, https://earthjustice.org/wp-content/uploads/energy_bomb_bitcoin_white_paper_101322.pdf (accessed Jul. 10, 2023).

[45] https://www.nytimes.com/2023/04/09/business/bitcoin-mining-electricity-pollution.html.

[46] https://ssrn.com/abstract=3779720.

[47] "Climate and Energy Implications of Crypto-Assets in the United States," White House Office of Science and Technology Policy, September 8, 2022, https://www.whitehouse.gov/wp-content/uploads/2022/09/09-2022-Crypto-Assets-and-Climate-Report.pdf.

[48] Nic Carter, "How Much Energy Does Bitcoin Actually Consume?," *Harvard Business Review*, May 5, 2021, https://hbr.org/2021/05/how-much-energy-does-bitcoin-actually-consume (accessed Jul. 10, 2023).

[49] Aaron Clark, "Ethereum's Energy Revamp Is No Guarantee of Global Climate Gains," Bloomberg, December 6, 2022, https://www.bloomberg.com/news/articles/2022-12-06/ethereum-cut-its-energy-use-99-but-climate-gains-may-be-curbed (accessed Jul. 10, 2023).

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Crime

Another concern for Brown were criminal activities facilitated by bitcoin and other digital assets. He had been watching a recent Senate hearing in which US Senator Elizabeth Warren made the following statement:

> So, big time financial criminals love crypto. Just last year, just in one year, crypto was the payment method of choice for international drug traffickers who raked in over a billion dollars through crypto. North Korean hackers, who stole $1.7 billion and funneled that money into their nuclear program. And ransomware attackers who took in almost $500 million.[50]

Indeed, the use of crypto in criminal activities, defined as funds sent to sanctioned wallets, had been increasing from less than USD5 billion in 2017 to over USD20 billion in 2022. For example, in May 2021, a Russian criminal group hacked Colonial Pipeline's internal computer systems, leading to gas shortages. Colonial Pipeline paid the group a ransom of USD4.4 million in crypto. In some cases, the public nature of a blockchain also helped to track down criminals. In the case of Colonial Pipeline, investigators eventually recovered part of the ransom.

Criminal activity worried Brown for two reasons. First, he was concerned that it might impede the adoption of bitcoin by other investors. Would traditionally conservative investors like pension funds and mutual funds buy bitcoin if it facilitated crime? Second, it was possible (likely, perhaps) that Congress would take steps to further regulate crypto markets in response to criminal activity. What form would such regulation take? Would it specifically target criminal activity or attempt to limit the growth of crypto more generally? Some have even called for Congress to ban crypto, something that could be disastrous for Brown's portfolio.[51]

Currency or Commodity?

Brown also needed to revisit what gave bitcoin value in the first place. (See the **Appendix** for a discussion of approaches that have been proposed to value bitcoin.) Ultimately, bitcoin was simply an entry in a digital ledger—just numbers on a screen. What made these numbers so valuable? He had often heard that bitcoin could be thought of as a currency or a commodity. He considered both arguments.

Bitcoin as a currency

Satoshi Nakamoto's original vision for bitcoin was that of "electronic cash."[52] Indeed, bitcoin shared some characteristics with widely used currencies. Similar to "fiat" currencies like the US dollar, bitcoin had no intrinsic value; however, bitcoin and US dollars differed in their use as a medium of exchange, a defining characteristic of currencies. The dollar was special because the US government deemed it legal tender, meaning that it must be accepted as payments for debts and other obligations. In practice, dollars were widely used to purchase goods and services, at least in the United States. Bitcoin was not. While some merchants accepted it for payments (e.g., Overstock.com and the Dallas Mavericks), the total volume sent through crypto payment

[50] "At Hearing, Warren Raises Concerns that Criminals, Terrorists, Rogue States Use Crypto to Launder Billions," Elizabeth Warren press release, February 14, 2023, https://www.warren.senate.gov/newsroom/press-releases/at-hearing-warren-raises-concerns-that-criminals-terrorists-rogue-states-use-crypto-to-launder-billions (accessed Jul. 10, 2023).

[51] Charlie Munger, "Why America Should Ban Crypto," *Wall Street Journal*, February 1, 2023, https://www.wsj.com/articles/why-america-should-ban-crypto-regulation-economy-finance-china-england-trading-currency-securities-commodity-gamble-11675287477 (accessed Jul. 10, 2023).

[52] https://bitcoin.org/bitcoin.pdf.

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processors paled in comparison to credit cards (approximately USD4 billion versus USD3.7 trillion in 2019).[53] Famously, even some bitcoin conferences did not accept bitcoin as payment for tickets.[54]

Two main factors explained bitcoin's failure to achieve widespread adoption for payments. First, its volatility was extremely high, roughly ten times that of major exchange rates (e.g., euros to US dollars).[55] Thus, bitcoin's purchasing power fluctuated dramatically from one day to the next, limiting its usefulness as a medium of exchange. Second, transactions on the bitcoin blockchain were slow. New blocks were mined approximately every 10 minutes, and transactions were typically not considered finalized until confirmed by multiple subsequent blocks.

Bitcoin's ability to serve as a currency was put to the test in El Salvador. In September 2021, President Nayib Bukele announced plans to make bitcoin legal tender, meaning that both the government and merchants were required to accept it as a form of payment. The motivation for doing so was to bring "more Salvadorans, about 70% of whom don't have bank accounts, into the formal economy."[56] The government encouraged citizens to download an app, Chivo, which could hold bitcoin and connect to local banks. Chivo helped to overcome bitcoin's slow transaction times by using a payment channel (the "Lightning Network") that settled transactions on a separate network before sending them to the bitcoin blockchain. The International Monetary Fund (IMF) raised concerns regarding the use of bitcoin as legal tender, arguing that it posed risks to financial stability and consumer protection.[57] Alvarez et al. found that the use of bitcoin in El Salvador was low, with less than 5% of sales across the country in bitcoin.[58]

Bitcoin as a commodity

Brown also considered the argument that, rather than a currency, bitcoin was a commodity akin to gold. To understand this comparison, it was useful to think about why gold was valuable. Gold was conductive and resistant to corrosion, giving it a wide range of electronic applications (e.g., semiconductor manufacturing). It was shiny and malleable, making it a popular choice for jewelry making; however, Urban J. Jermann argued that these uses of gold only accounted for about a third of its value; the bulk of its value derived from its use as an investment asset.[59] Specifically, investors perceived gold as a store of value, an asset that maintained its purchasing power into the future—a particularly attractive investment in periods with low or negative real interest rates. Like gold, much of bitcoin's value emanated from the fact that people agreed it was valuable. In addition, both were scarce and could only be produced via costly mining operations. Finally, both were fungible, meaning that a unit of either bitcoin or gold could be replaced with an equivalent unit.

But there were also important differences between gold and bitcoin. For one, gold's history as a store of value stretched back millennia. Bitcoin did not exist until 2009. In addition, gold had actual uses; Bitcoin did not. Jermann noted that the use of gold for jewelry or as a productive input essentially provided a (positive)

[53] Leigh Cuen, "Bitcoin Usage Among Merchants Is Up, According to Data from Coinbase and BitPay," CoinDesk, September 13, 2021, "https://www.coindesk.com/business/2020/02/03/bitcoin-usage-among-merchants-is-up-according-to-data-from-coinbase-and-bitpay/ (accessed Jul. 10, 2023).

[54] Saheli Roy Choudhury, "A Bitcoin Conference Has Stopped Taking Bitcoin Payments Because They Don't Work Well Enough," CNBC, January 10, 2018, https://www.cnbc.com/2018/01/10/bitcoin-conference-stops-accepting-cryptocurrency-payments.html (accessed Jul. 10, 2023).

[55] Dirk G. Baur and Thomas Dimpfl, "The Volatility of Bitcoin and Its Role as a Medium of Exchange and a Store of Value," *Empirical Economics* 61 (2021), https://doi.org/10.1007/s00181-020-01990-5 (accessed Jul. 10, 2023).

[56] Oscar Lopez and Ephrat Livni, "In Global First, El Salvador Adopts Bitcoin as Currency," *New York Times*, September 7, 2021, https://www.nytimes.com/2021/09/07/world/americas/el-salvador-bitcoin.html (accessed Jul. 10, 2023).

[57] "IMF Executive Board Concludes 2021 Article IV Consultation with El Salvador," IMF, January 25, 2022, https://www.imf.org/en/News/Articles/2022/01/25/pr2213-el-salvador-imf-executive-board-concludes-2021-article-iv-consultation (accessed Jul. 10, 2023).

[58] Fernando E. Alvarez, David Argente, and Diana Van Patten, "Are Cryptocurrencies Currencies? Bitcoin as Legal Tender in El Salvador," National Bureau of Economic Research working paper no. 29968, https://www.nber.org/system/files/working_papers/w29968/w29968.pdf (accessed Jul. 10, 2023).

[59] https://www.nber.org/system/files/working_papers/w29968/w29968.pdf.

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floor on its price (i.e., an embedded put option). Bitcoin's intrinsic value was zero. Nevertheless, bitcoin had some advantages over gold as a store of value. For one, bitcoin's storage costs were close to zero. It was also relatively easy to send a large amount of bitcoin to another person. Sending an equivalent value of gold might entail considerable expense.

Theoretical arguments aside, Brown noticed that the prices of bitcoin and gold behaved quite differently. Not only was bitcoin more volatile, but it often did not move in the same direction as gold. At least recently, bitcoin seemed to have a stronger correlation with tech stocks (**Exhibit 7**).

Decision Time

Brown took a deep breath and tried to organize his thoughts. The recent crypto winter had taken its toll on both him and his investors. And yet, despite its recent performance, bitcoin was still at double its pre-COVID-19 price. Could a return to those levels only be a matter of time? If so, the LPs pressuring him were right, and he should cut the position or even short bitcoin. On the flip side, Brown had been in this situation before, with investors pushing him to sell after bitcoin crashed in 2018. Brown had resisted such calls then and was eventually vindicated. Perhaps history would repeat itself.

To hodl or not to hodl? That was the question.

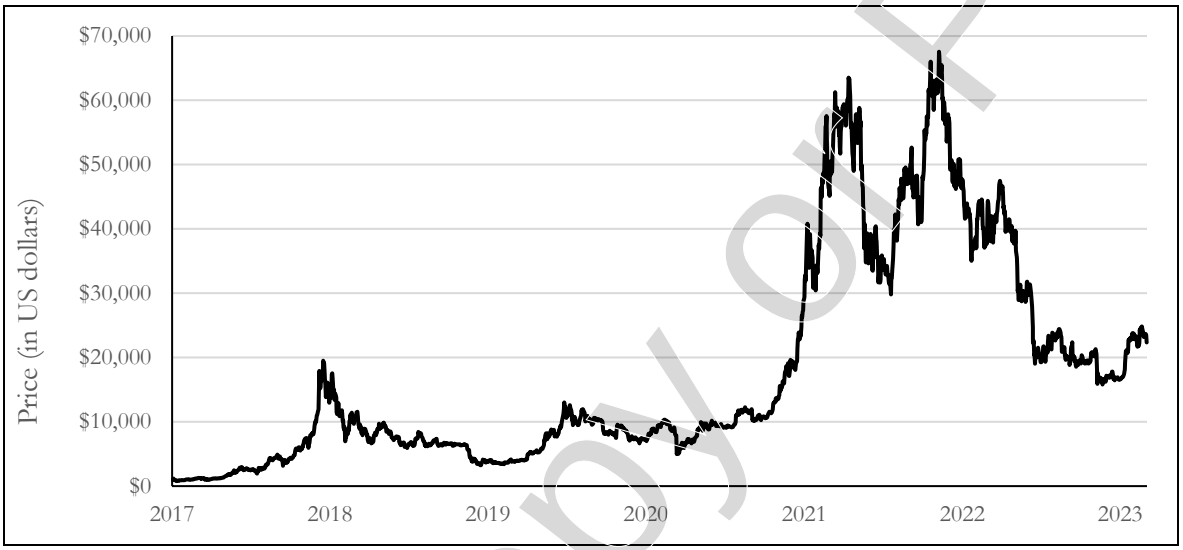
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Exhibit 1

Bitcoin 2023: Keep "Hodling"?

Bitcoin Price (January 1, 2017, to March 3, 2023)

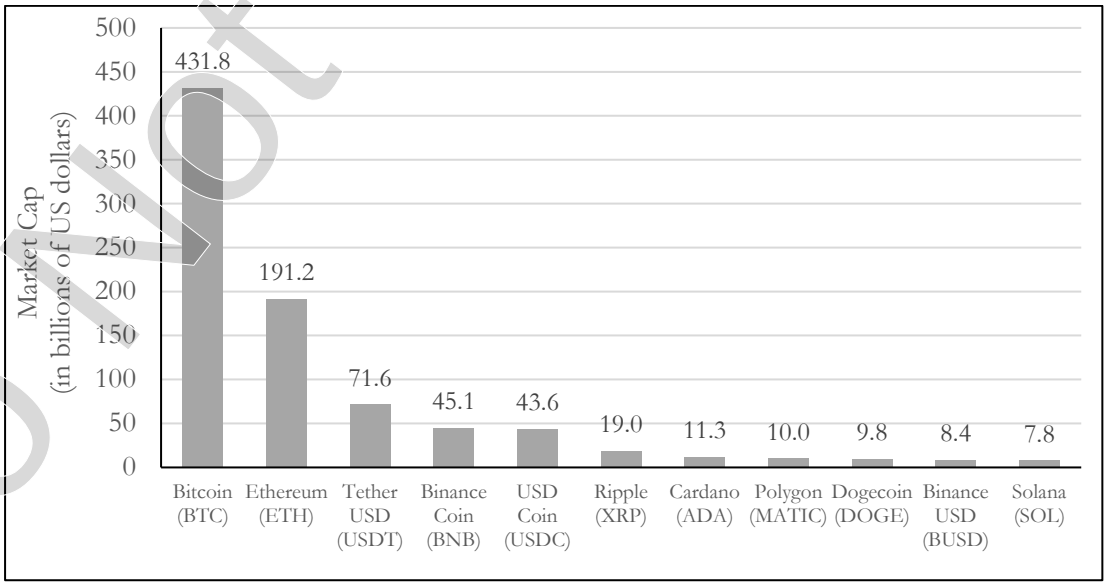


Data source: Yahoo! Finance.

Exhibit 2

Bitcoin 2023: Keep "Hodling"?

Largest Cryptocurrencies by Market Capitalization (March 2023)



Note: USDT, USDC, and BUSD were stablecoins (cryptocurrencies pegged to the US dollar).

Data source: CoinMarketCap.

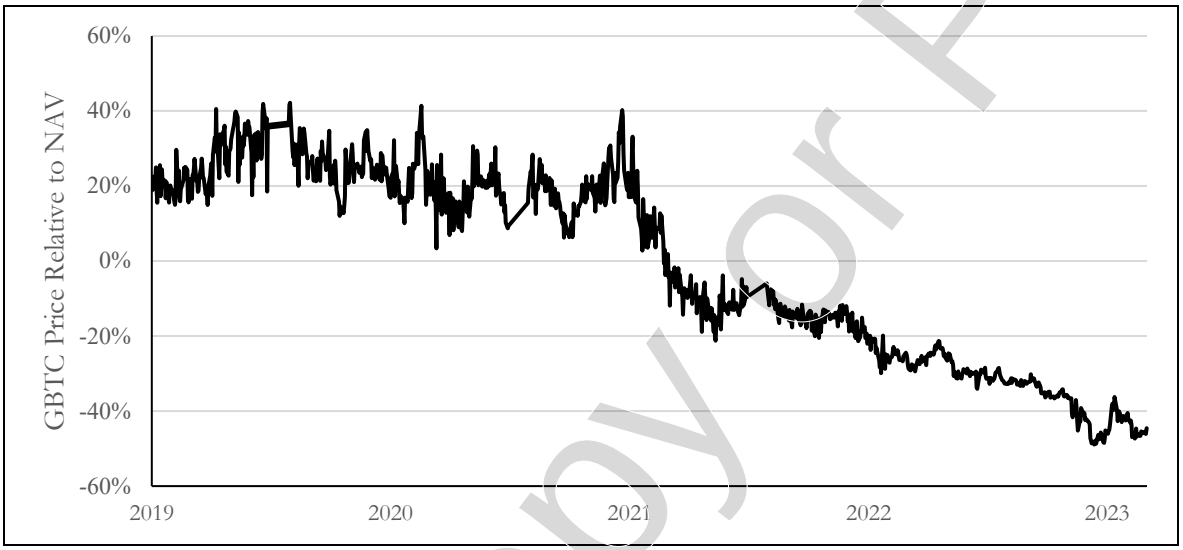
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Exhibit 3

Bitcoin 2023: Keep "Hodling"?

Grayscale Bitcoin Trust Premium to NAV (January 2, 2019, to March 3, 2023)

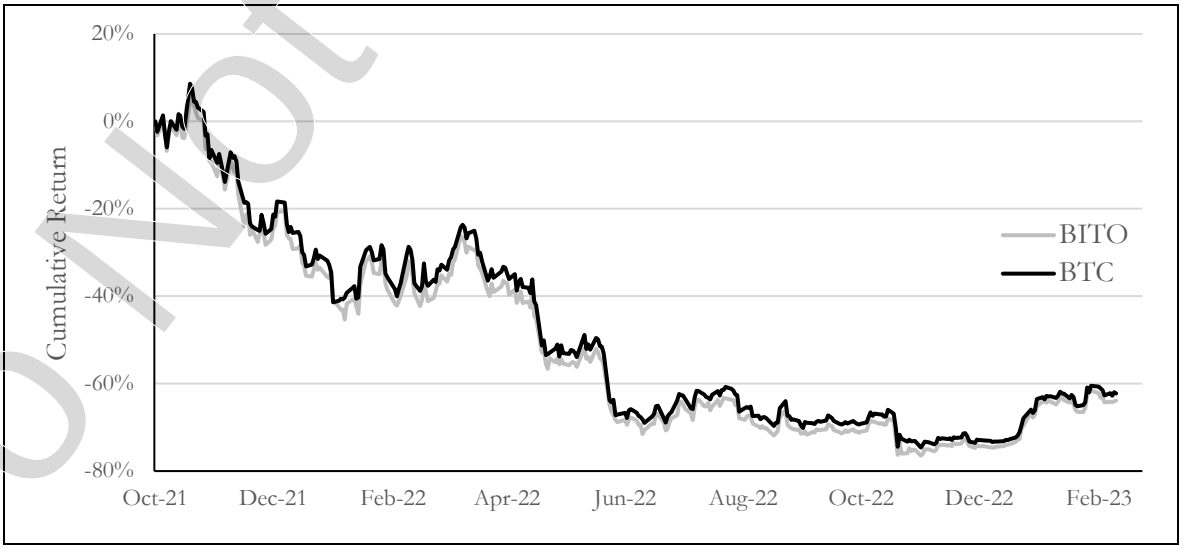


Sources: Grayscale Investments and authors' calculations.

Exhibit 4

Bitcoin 2023: Keep "Hodling"?

BTC versus BITO Returns (October 21, 2021, to March 3, 2023)



BTC = bitcoin; BITO = ProShares Bitcoin Strategy ETF.

Data source: Yahoo! Finance.

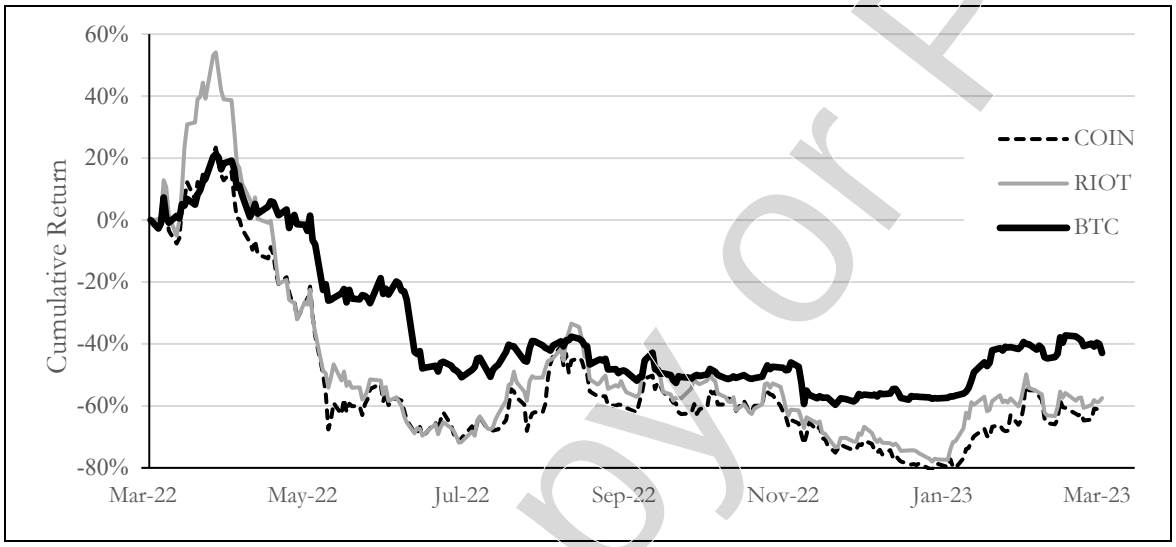
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Exhibit 5

Bitcoin 2023: Keep "Hodling"?

BTC versus Crypto Equity Returns (March 3, 2022, to March 3, 2023)



COIN = Coinbase; RIOT = Riot Platforms; BTC = bitcoin.

Data sources: Grayscale Investments and authors' calculations.

Exhibit 6

Bitcoin 2023: Keep "Hodling"?

Annualized Bitcoin Mining Energy Consumption (January 2017 to March 2023)



Data source: University of Cambridge Centre for Alternative Finance.

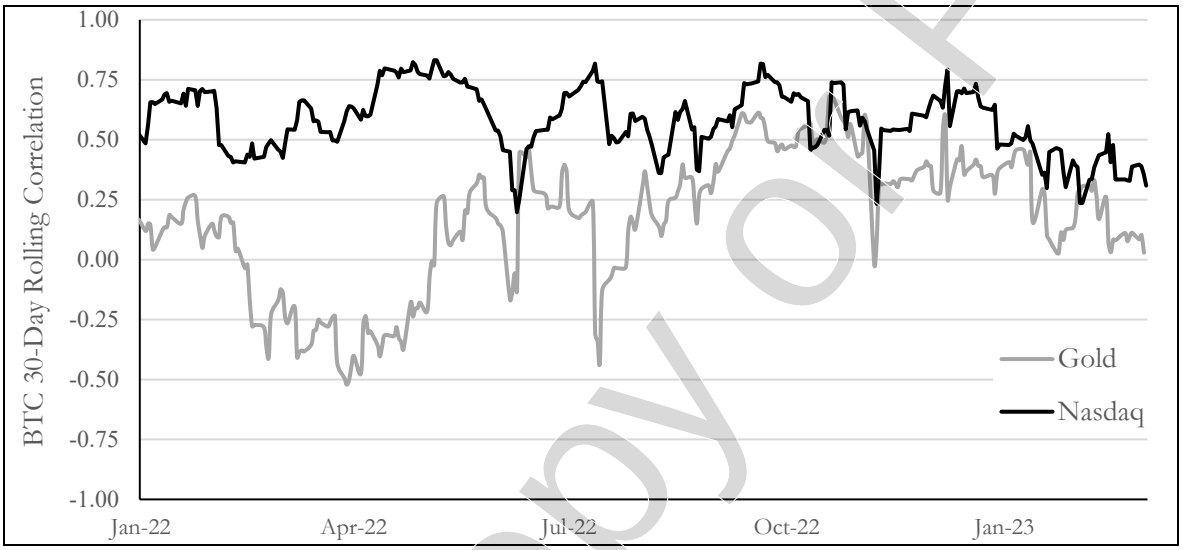
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Exhibit 7

Bitcoin 2023: Keep "Hodling"?

30-Day Rolling Correlation of Bitcoin with Gold and NASDAQ 100



Note: Gold returns measured using GLD, a Standard & Poor's Depository Receipt (SPDR) exchange-traded fund (ETF) for gold shares. NASDAQ 100 returns measured using the Invesco QQQ Trust.

Data sources: Yahoo! Finance and authors' analysis.

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Appendix

Bitcoin 2023: Keep "Hodling"?

Valuing Bitcoin

This appendix provides a brief discussion of common valuation frameworks for bitcoin. Valuing bitcoin presents unique challenges relative to other financial assets. Typically, assets are valued by discounting their expected future cash flows. For example, the Gordon Growth model, a staple of introductory finance classes, says that the value of a stock is equal to the present value of its future dividends. Bitcoin does not generate cash flows and thus cannot be valued using such a model. Of course, one could argue that in the absence of cash flows, the value of a bitcoin should be zero. Market prices indicate that many disagree with this view.

a. Relative valuation

One way to value bitcoin is to compare its market cap to that of alternative asset classes. Hal Finney, a computer scientist who received the first bitcoin transaction from Satoshi Nakamoto, famously proposed such an approach in January 2009, just a week after the first bitcoin block was mined:

> One immediate problem with any new currency is how to value it. Even ignoring the practical problem that virtually no one will accept it at first, there is still a difficulty in coming up with a reasonable argument in favor of a particular non-zero value for the coins.
>
> As an amusing thought experiment, imagine that Bitcoin is successful and becomes the dominant payment system in use throughout the world. Then the total value of the currency should be equal to the total value of all the wealth in the world. Current estimates of total worldwide household wealth that I have found range from $100 trillion to $300 trillion. With 20 million coins, that gives each coin a value of about $10 million.[1]

While intentionally outlandish, Finney's logic has commonly been used in the context of "digital gold." Harvey et al. note that the value of mined and unmined gold is approximately USD15 trillion.[2] To the extent that bitcoin displaces some demand for gold, one could argue that bitcoin should partially account for this value. For example, perhaps the gold/bitcoin split should be 50/50, which would imply a market cap of bitcoin of USD7.5 trillion, or about USD350,000 per coin. A key challenge is that it is exceedingly difficult to argue what the bitcoin market share should be. Why 50% and not 5% or 0.5%? Some even argue that bitcoin is "better at being gold than gold" and should have a higher market cap.[3]

b. Metcalfe's Law

Another approach that has been used to value bitcoin is Metcalfe's Law, perhaps more appropriately called a theory, which expressed the relationship between the size of a network and its value. The key idea was that the number of connections in a network increase at a faster rate than the number of new users. Consider a social media network like Facebook. If a network has 3 members, there are a total of 3 connections between members. If the network has 4 members, there are now 6 connections.

---

[1] Hal Finney, "Re: Bitcoin v0.1 Released," Mail Archive, January 11, 2009, https://www.mail-archive.com/cryptography@metzdowd.com/msg10152.html (accessed Jul. 10, 2023).

[2] Campbell R. Harvey, Tarek Abou Zeid, Teun Draaisma, Martin Luk, Henry Neville, Andre Rzym, and Otto van Hemert, "An Investor's Guide to Crypto," September 1, 2022, https://ssrn.com/abstract=4124576 (accessed Jul. 10, 2023).

[3] Tyler Winklevoss, "The Case for $500K Bitcoin," Winklevoss Capital, August 27, 2020, https://winklevosscapital.com/the-case-for-500k-bitcoin/ (accessed Jul. 10, 2023).

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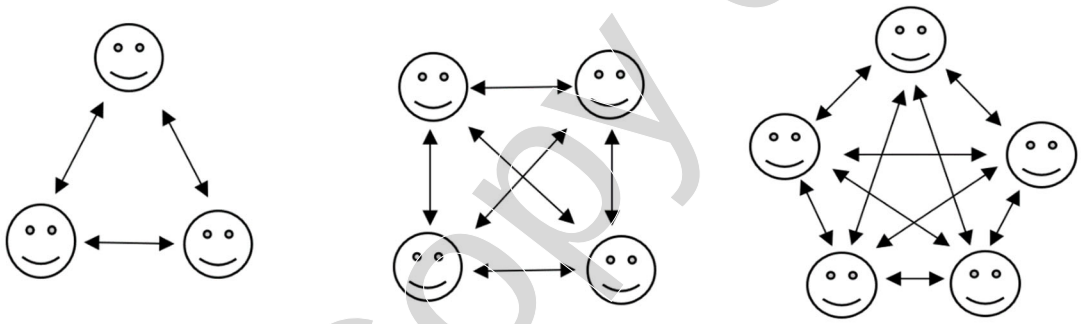
Appendix (continued)

Valuing Bitcoin

Notice that the number of connections increased at a faster rate than the number of members of the network. As the number of members grows, the size of the network continues to grow at a faster rate. For example, if there are 5 members in the network, the number of connections is 10. See **Figure 1**.

Figure 1: Metcalfe's Law.

Network with:



Source: Created by authors.

Metcalfe's Law says that the value of a network is proportional to the square of the number of users. More formally, $Value = \alpha \times n^2$, where n = the number of users and α is a constant. Thus adding new members to a network will have increasing returns: each new connection makes the network more valuable than the one before it. How well does it work for crypto? The evidence is somewhat mixed. Harvey et al. note that Metcalfe's Law is useful for describing the valuation of developed cryptocurrencies, namely bitcoin and ether, but not less-developed ones. Moreover, even if the model is correct, the value of the constant (α) is unknown and could change over time. Harvey et al. conclude that while Metcalfe's Law potentially provides a useful conceptual framework, it is insufficient for valuation purposes.[4]

c. Stock-to-flow model

A final valuation approach that has received considerable attention (on social media, in particular) is the stock-to-flow model, proposed by an anonymous Twitter user.[5] The model values bitcoin in terms of the inflation in the supply of coins resulting from miner rewards. Inflation decreases over time, eventually converging to zero around the year 2140.[6] The model assumes that as inflation decreases, the price will increase. This framework suffers from several conceptual flaws. Most glaringly, Harvey et al. note that it implies an infinite price for bitcoin once the flow of new coins is zero.

[4] https://ssrn.com/abstract=4124576.

[5] PlanB [pseud.], "Modeling Bitcoin Value with Scarcity," *Medium*, March 22, 2019, https://medium.com/@100trillionUSD/modeling-bitcoins-value-with-scarcity-91fa0fc03e25 (accessed Jul. 10, 2023).

[6] "What Happens When All 21 Million Bitcoin Are Mined?" NASDAQ, June 8, 2022, https://www.nasdaq.com/articles/what-happens-when-all-21-million-bitcoin-are-mined (accessed Jul. 10, 2023).

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